Chimdiume Azie

AI/ML Developer · B.Sc. Computer Science

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Profile

Recent Computer Science graduate (Godfrey Okoye University, 2026) focused on building explainable AI systems end to end. Built a deep learning system that classifies Alzheimer's stages from brain MRI scans and explains its predictions, together with the hospital web application around it, built to run on a laptop with no GPU. Also building Nellow, an AI agent that pairs a local Llama model with Claude as a cloud reasoning model. Seeking an entry-level AI/ML or software engineering role.

Projects

Alzheimer's Disease Detection System using Deep Learning and Explainable AI (Final-year project, github.com/Achimdi/neuroscan-ai)

- Trained an EfficientNet-B0 classifier (PyTorch, transfer learning) to sort brain MRI scans into four stages (NonDemented, VeryMildDemented, MildDemented, ModerateDemented) using the OASIS MRI dataset.
- Added Grad-CAM heatmaps that show which brain regions drive each prediction; for Mild and Moderate cases they concentrated on the hippocampus, medial temporal lobe and ventricles.
- Built a Flask hospital web application with Admin and Doctor roles (JWT authentication, role-based access, one-time registration codes), patient records, MRI upload with predicted stage, confidence score and heatmap, saved result history, and an audit trail for patient transfers (SQLite, SQLAlchemy).
- Handled class imbalance with a class-weighted loss, and trained and ran the whole system on a laptop with no GPU to suit resource-limited hospitals.
- Results: 98.66% validation accuracy. Accuracy on the held-out test set was 77.35%; I traced the gap to a skewed original split (88.5% NonDemented in the test set against 40.8% in training) and documented stratified splitting and clinical validation as next steps. Research prototype, not clinically validated.

Nellow: AI Agent (github.com/Achimdi/nellow)

- Built an AI agent that searches the live web, reads what it finds and answers with sources, and can run on my own hardware or in the cloud.
- Hybrid two-model design: a Llama model runs locally on my own machine, at no per-request cost, for quick answers that need little reasoning, and Claude is used as the cloud reasoning model for deeper, multi-step thinking.
- Web workspace with chat history, projects, a speed/model selector, file attachments, and voice input and spoken output controls. Hosting is planned.

Skills

- **Machine learning and AI:** PyTorch, transfer learning, CNNs, Grad-CAM and explainable AI, scikit-learn, OpenCV, NumPy, Pandas, model evaluation (precision, recall, F1, confusion matrix, ROC)
- **Web and backend:** HTML, CSS, JavaScript, Flask, REST APIs, JWT authentication, SQLite and SQLAlchemy
- **Mobile:** Flutter (working knowledge)
- **Tools:** Python, Git and GitHub, VS Code

Education

Bachelor of Science (B.Sc.) in Computer Science, Godfrey Okoye University, 2026

Relevant coursework: Expert Systems, Computer Networks and Communication, Compiler Construction, Structured Programming, Human Computer Interaction, Systems Analysis and Design

References

Available on request.